

AN FSM BASED ELEVATOR SYSTEM

IMPLEMENTED ON FPGA

Group 07

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OUTLINE

Features

State Machine

An Elevator System

State Encoding

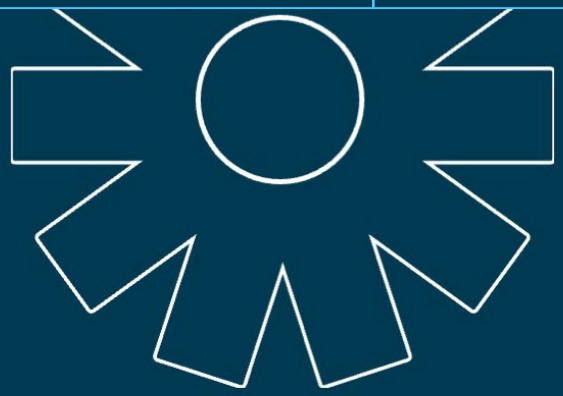
Feedback Drive

Next State Logic

Input Handling

FPGA Implementation

Limitations of the Implementation

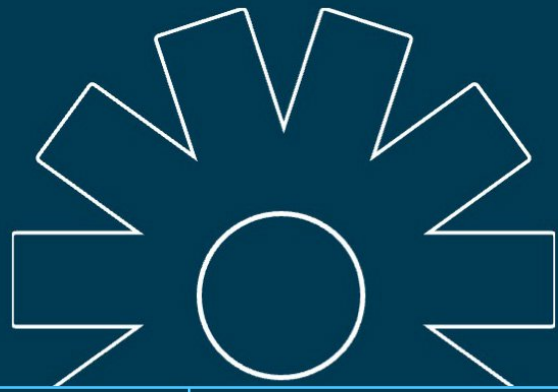


FEATURES

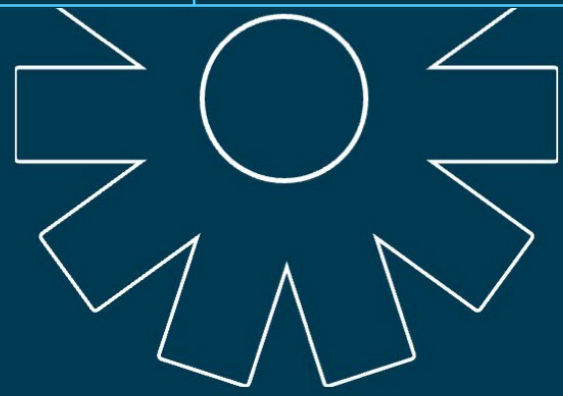
- Four floor system
- Multiple request handling
- Resolution of simultaneous requests
- All the call, floor, door buttons
- Efficient request fulfilment
- An instructive tool on FSM
- Shows correct floor reading unlike *that* ECE lift

STATE MACHINE

- States
- Transitions
- Inputs
- Outputs
- Initial State



- Floors
- Next State Logic
- Requests/Calls, Door
- Direction, Door, Position
- Reset

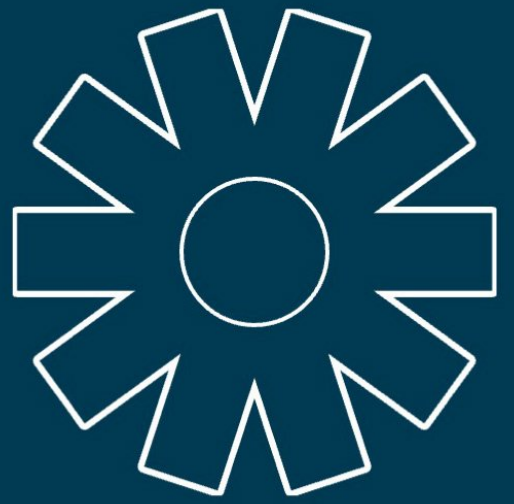


AN ELEVATOR SYSTEM

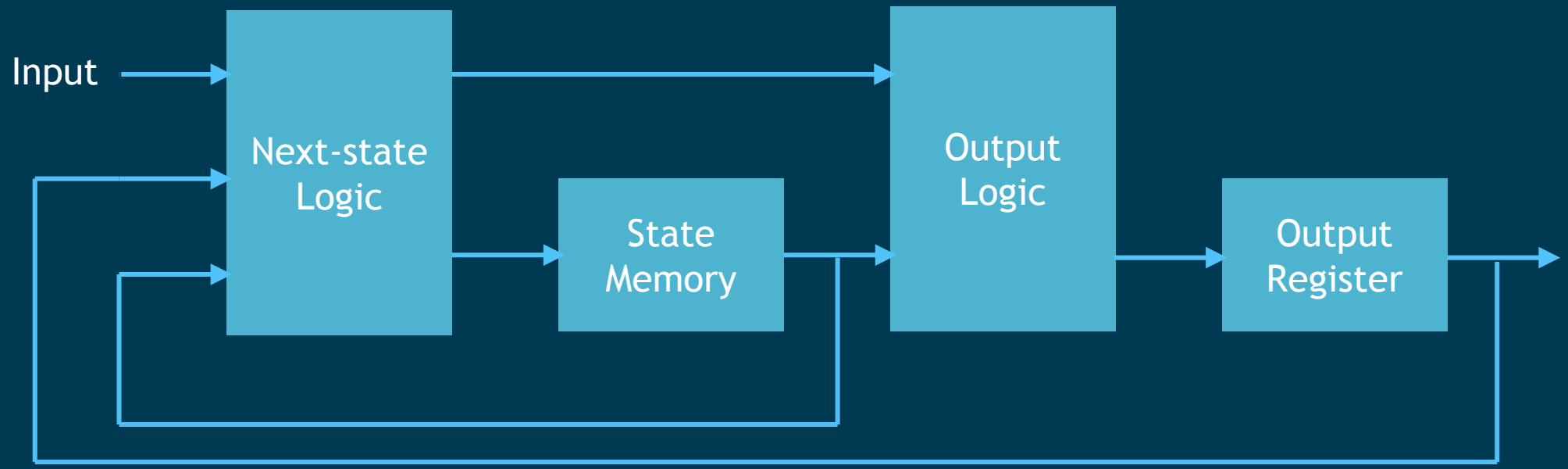
STATE ENCODING

A Mealy state machine implementation requires 4 states.

Ground	00
First	01
Second	10
Third	11



FEEDBACK DRIVE

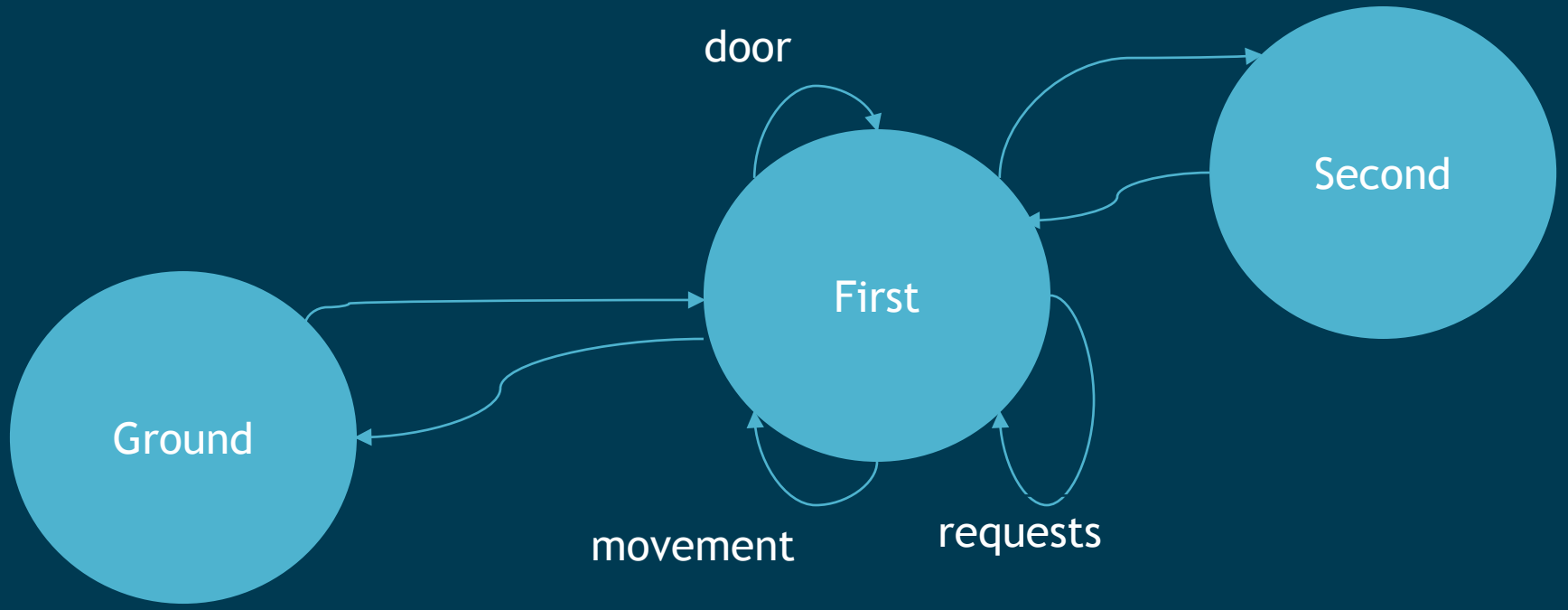


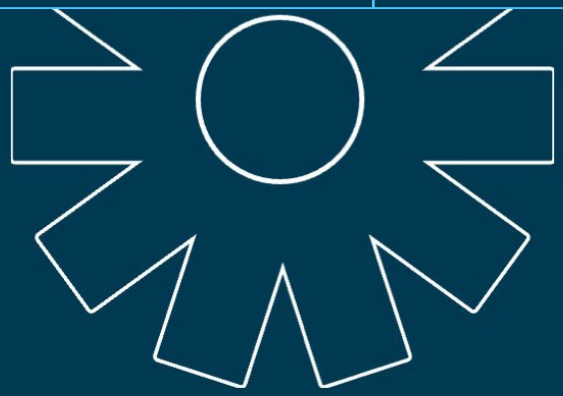
NEXT STATE LOGIC

- Inputs are stored
- Minimizes time and distance
- Processes input requests and current outputs
- Casework
- State transitions at each clock cycle

- What's the challenge?
  - Accounting for outputs
- What if there's a conflict?
  - Inside requests are prioritized

STATE TRANSITION





INPUT HANDLING

- Arrays to store inputs
- Multiple simultaneous requests
- Request clearing
- Door logic

- Manual clock
- Switches for input
  - Buttons for “up” requests
  - “Down” requests
  - “Floor” requests
  - Door operation
- Reset button

- Visual outputs of the elevator
  - Current floor on 7 segment display
  - Current direction
- Further
  - LED to indicate door status
  - LED to indicate movement status

FPGA IMPLEMENTATION

LIMITATIONS OF THE IMPLEMENTATION

Manual Clock

Implementation possible with automatic clock and time delays

Push Buttons

Should be replaced with reliable pulse switches

Extension

The FPGA has limited possibility to extend to many floors

THANK YOU

FOR ANY QUERIES

USE INSTITUTIONAL E-MAIL

FOR ANY OF THE FOLLOWING IDS

2006010, 2006024, 2006026, 2006031